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UGV–Based Multi–Spectral Detection of Surface & Shallow–Buried Landmines

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Chapter One : Introduction

1-1 Project Over View and Background :

Landmines have caused serious threats to human lives in many regions of the world. Humans are extremely exposed to risk when detecting landmines manually. This is a very time consuming and tedious process. In order to reduce the risks of humans getting injured or killed robotic systems have been designed to do this.

This project focuses on creating an autonomous landmine detection UGV.
It consists of a robust mobile robot platform along with multiple sensor integrations and modern localization and mapping systems. Using modern robotics techniques this UGV can be used to traverse dangerous environments while detecting any potential explosive hazards and mapping their position.

Besides the issue of autonomous navigation, the ability to reliably detect hazards is also a problem when using traditional metal detection and monochrome RGB cameras due to the high rate of false positives. Many modern Anti-Personnel and Anti-Tank Landmines are designed to have little to no metal and are often hidden in complex terrain. This project addresses this problem through incorporating Artificial Intelligence into the perception pipeline. Using multi-spectral and thermal computer vision, the different wavelengths of light from three different sources can be captured; Visible (RGB), Near-Infrared and Long Wave Infra-Red. This allows the onboard deep learning algorithm to classify targets and create a hazard map based on the visual features of the target and thermal anomalies due to heat from movement of the soil and differing thermal inertia.

1-2 Problem Statement and Practical/Academic Significance :

Manual landmine detection in post conflict or contaminated environments has a high

risk associated with it. Humans have to go into dangerous areas to detect landmines making the situation very slow and dangerous.

Furthermore, conventional electromagnetic detection methods suffer from excessive false-positive rates in battle-contaminated terrains saturated with metallic shrapnel, while completely failing to detect low-metal or plastic-encased explosive devices.

Academic and Practical Significance:

- **Safety Enhancement:** Removes the human from the danger zone completely and uses an autonomous or tele-operated robot instead
- **Efficiency and Precision:** Uses automated scanning and detection methods using advanced path planning algorithms to increase the detection rate.
- **Technological Integration:** Integrates multiple sensors with ROS2 navigation systems to validate they can work together in complex environments
- **Multi-Spectral AI Perception:** We used deep Convolutional Neural Networks to detect real explosive threats from environmental clutter using multi-modal RGB, NIR and Thermal IR sensors. This reduced false alarm rates.
- **Edge-AI and Robotics:** We demonstrated how deep learning inference can be used in real time with ROS 2 navigation nodes. It allowed us to apply high accuracy computer vision to our low powered robotic edge devices.

1-3 Main and Sub Project Objectives :

Design, model and simulate an autonomous ground vehicle that can navigate unknown or semi structured environments. It uses a multi-spectral deep

learning perception system to identify, detect and accurately map surface and shallow buried landmine hazards.

Sub-Objectives / Technical Requirements:

- **Multi Spectral Data Gathering and Preprocessing:** Capture, align and augment multi-spectral datasets. These will be composed of RGB, NIR, Thermal/LWIR images of surface and shallow buried landmines.
- **Deep Learning Model and Sensor Fusion. Train:** tune and accurately test real-time object detection models. This will include models such as multi stream YOLO models that use either feature level or decision level sensor fusion to provide the best possible recall and precision.
- **AI Perception using ROS 2:** Deploy the trained deep learning model within a dedicated ROS 2 perception node that subscribes to simulated camera streams .
- **Mobility and Chassis Design:** Design a 4WD or differential-drive mobile platform capable of traversing rough outdoor terrains typical of minefields.
- **Navigation and Mapping (SLAM):** Implement Simultaneous Localization and Mapping (SLAM) using LiDAR and odometry sensors within ROS 2 Gazebo simulation environments.
- **Sensor Payload Integration:** Integrate simulated metal detection and proximity sensing modules to identify buried or surface-level threats.
- **Autonomous Path Planning:** To create path planning algorithms that can fully scan an area. Performance Metrics: High detection accuracy, stable trajectory tracking and real time data transmission.

1-4 Basic Definitions and Concepts:

- **Autonomous Ground Vehicle (UGV):** A vehicle that operates while in contact with the ground and without direct human intervention, utilizing onboard sensors and computational intelligence.
- **SLAM (Simultaneous Localization and Mapping):** The computational problem of constructing or updating a map of an unknown environment while simultaneously keeping track of an agent's location within it.
- **ROS 2 (Robot Operating System):** A flexible framework for writing robot software, providing services designed across a diverse cluster, including hardware abstraction, low-level device control, and message-passing implementations.
- **Coverage Path Planning (CPP):** The task of determining a path that passes through every point of a given area or environment while avoiding obstacles.

- **Multi-Spectral & Thermal Imaging:** Imaging using specific wavelengths of the electromagnetic spectrum. This can include visible light (RGB), Near-Infrared (NIR) and Long-Wave Infrared (LWIR). This allows different materials to be identified based on their reflectance 'signature' and thermal anomalies below the surface of materials.

- **Multi-Modal Sensor Fusion:** Combining information from multiple sources together, including optical and thermal cameras. This can happen at pixel, feature or decision level, to give a more accurate, robust and complete picture than would otherwise be possible with one single sensor.

- **Deep Learning Object Detection:** Using Deep Convolutional Neural Networks (CNNs) or Vision Transformers to detect and classify multiple objects of interest

in an image. This will also provide you with bounding box coordinates and confidence scores for the detected objects.

1-5 Project Scope and Technical Requirements :

- **Project Scope:** Designing a CAD Model, Power Budget, Kinematic Mathematical Model, ROS 2 Jazzy Software Architecture, Gazebo Simulation and Testing the performance in different terrains. However, the project does not include manufacturing the robot in mass production and deploying to live mines. AI work includes pre-processing of multi-spectral data, training deep learning models, hyperparameter tuning, multi-modal fusion assessment and integration into real-time AI using ROS 2 perception nodes. However, no real-time re-training of models on devices during missions is included.

- **Technical Requirements:** The simulated vehicle must maintain a steady cruising speed, withstand nominal slope inclines, provide real-time sensor feedback, and accurately map hazard locations

  In addition, the AI perception pipeline must meet a desired Recall value of $\geq 95\%$ (ensuring safety by reducing explosive hazard misses), Mean Average Precision $\geq 85\%$ and real time inference throughput of ≥ 15 FPS to prevent bottlenecks in the autonomous navigation control loop.

1-6 Project Contribution to Knowledge and Field :

This project contributes to the field of humanitarian de-mining robotics by proposing an integrated simulation framework that combines autonomous coverage path planning with multi-sensor anomaly detection. It provides a reliable testing testbed for

developing low-cost modular UGVs deployable in hazardous reconstruction zones, bridging the gap between theoretical control algorithms and practical field applications.

This work also provides a comparison of multi-spectral fusion methods against single-modality counterparts for the purpose of surface and shallow-buried target detection and identification. It incorporates real-time deep learning-based anomaly detection with ROS 2 spatial mapping to form a complete perception-to-action feedback loop. In this system, detected spectral anomalies are used to update costmaps for enabling autonomous navigation.

1-7 Adopted Design Methodology :

The project follows a systematic engineering design methodology consisting of four main phases :

1. **Requirement Analysis and Literature Review:** Defining mechanical and computational functional specifications, analyzing existing robotic landmine detection systems, and reviewing state-of-the-art multi-spectral deep learning architectures and public minefield datasets.
2. **Modeling and System Design:** Formulating the kinematic equations, designing the mechanical CAD assembly, selecting virtual sensors, curating and augmenting the multi-spectral/thermal image dataset, and designing the neural network fusion architecture
3. **Simulation and Implementation:** Training and optimizing the deep learning detection models, constructing the virtual world in Gazebo, writing ROS 2 navigation and AI inference nodes in Python, and executing integrated navigation and spectral detection scenarios.
4. **Testing, Evaluation, and Analysis:** Analyzing path coverage efficiency and localization drift alongside AI perception metrics—specifically Precision, Recall,

F1-score, inference latency, and spatial hazard-mapping accuracy under simulated environmental noise and obstacles

1-8 Report Structure :

This report is organized into five chapters:

- **Chapter 1** introduces the project background, problem statement, objectives, scope, and methodology.
- **Chapter 2** reviews related literature, existing robotic landmine detection systems, multi-spectral and thermal imaging principles, deep learning object detection architectures, multi-modal fusion techniques, and identified research gaps.
- **Chapter 3** details the complete system architecture, covering mechanical/electrical design, mathematical kinematic modeling, multi-spectral dataset preprocessing, deep learning model architecture, and the ROS 2 Gazebo simulation setup.
- **Chapter 4** presents the implementation details, deep learning training and validation metrics (Precision, Recall, and confusion matrices), simulation results, spatial mapping accuracy, and comparative performance analysis
- **Chapter 5** concludes the thesis, highlights challenges faced, and outlines recommendations for future developments.

1-9 Conclusion :

This chapter has provided the background of the project, motivation and clear objectives. The next chapter will look at the current literature and identify the research gap.

Chapter Two : Literature Review And Related Work

2-1 Related Work In The Project Domain :

Recent advancements in robotic humanitarian demining have shifted from heavy, tele-operated mechanical flails to lightweight, autonomous Unmanned Ground Vehicles (UGVs) capable of non-destructive mapping.

1. Four-Wheel Skid-Steer Demining Rovers (e.g., Sapper-UGV & Minesweeper Platforms): Recent studies on rough terrain demining platforms emphasize 4WD skid-steering and rocker-bogie suspension geometries to traverse ungraded post-conflict soil. These studies demonstrate that lightweight chassis designs reduce vertical ground pressure below the activation threshold of pressure-fused anti-personnel mines. However, many existing platforms rely on manual tele-operation or simple waypoint following, which leaves coverage gaps in unstructured fields.
2. SLAM and Autonomous Coverage Path Planning in Unstructured Environments: Contemporary mobile robotics literature leverages the Robot Operating System (ROS 2) alongside 2D/3D LiDAR Simultaneous Localization and Mapping (SLAM) and Inertial Measurement Unit (IMU) data. Studies evaluating back-and-forth Coverage Path Planning demonstrate up to 98% area coverage efficiency compared to 65–75% in manual human sweeps, while dynamically routing around surface obstacles.

   Concepts Adopted for This Project (Robotics): This project adopts a 4WD differential/skid-steer kinematic configuration simulated in Gazebo, paired with ROS 2 Jazzy. It utilizes LiDAR-based SLAM for real-time occupancy grid generation and implements systematic Boustrophedon Coverage Path Planning to ensure complete, overlap-controlled scanning of the target zone.

Alongside robotic navigation, research efforts have been focusing on improving the shortcomings of electro-magnetic metal detection systems, which struggle with high false positive rates at detecting mines with a high shrapnel content.

1. Thermal (LWIR) Anomalies: Recent remote sensing papers have shown that subsurface objects will change the natural heat flow in the ground. The thermal conductivity and thermal inertia (heat capacity) of a landmine casing will differ to surrounding land. This will cause a temperature gradient across the surface of the ground when heated by the sun. Studies using Long Wave Infra-Red cameras (LWIR) have used CNNs to successfully identify anomalies below the surface. However, single band thermal detection can be hindered by false positives from rocks and shadow from vegetation.
2. Multi-Spectral Detection (RGB + NIR + LWIR): Cutting edge perception research has shown using multi-spectral methods with RGB visible light, Near Infra-Red (NIR) and thermal. The single stage object detectors such as YOLO can be used along with multi-modal feature fusion backbones to identify objects. The RGB images will give the texture of the mine casing and tripwire. The NIR images will identify the disturbed land and stressed vegetation. The Thermal images will identify the subsurface heat anomalies.
3. Concepts Used in this Project : The RGB, NIR and Thermal IR Multi-Spectral Deep Learning pipeline has been used within this project. A fine-tuned YOLO based multi-modal architecture is used within ROS 2. The object detector will publish the 2D detections of each modality within the 3D global hazard co-ordinates of the robots navigation map.

2-2 Comparative Analysis of Current Systems :

The table below compares the various technical approaches to addressing landmine detection challenges, outlining the strengths and limitations of each system:

Study / System	Technologies & Sensors	Key Strengths	Limitations / Weaknesses
UAV-Based Detection (2020)	Aerial drones, optical/thermal payloads, aerial photogrammetry	High-speed aerial coverage of large areas; completely safe from triggering anti-personnel pressure mines	Incapable of detecting deeply buried landmines; highly vulnerable to weather and wind conditions
suland (2024)	Surface/subsurface navigation UGVs, proximity sensors, enhanced traction	Precise ground interaction; robust mobility in rugged, post-conflict terrains	Slower relative scanning speed; higher localized energy consumption
MineInsight (2025)	AI-driven anomaly classifiers, real-time sensor fusion, thermal/spatial mapping	High analytical accuracy; real-time interpretation of sensor streams reducing false positives	Relies heavily on clean, calibrated input data streams to maintain classification integrity